note0403

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Often, the number of genetic marker covariates (m) included in genomic models exceeds the number of records (n). When the genomic model is parameterised in terms of genetic marker effects that are the focus of interest, given an $m \gg n$ setup, some form of regularisation is needed in order to obtain a solution for the m genetic marker effects using the n records available.

Consider a standard genomic model where y, an $n \times 1$ column vector of data has zero mean. The hierarchical structure of the model is as follows:

$$y \mid \beta, \sigma^2 \sim N\left(X\beta,\, I\sigma^2\right), \tag{1a}$$

$$\beta \mid \sigma^2_\beta \sim N\left(0,\, I\sigma^2_\beta\right). \tag{1b}$$

Regularisation is introduced by assigning a distribution to the vector β. The column vector of marker effects β, the focus of inference, is of dimension $m \times 1$ and in a first stage, it is assumed that the variance components σ^2 and σ^2_β are known so that β is the only unknown parameter. Matrix X is often rank deficient: $\text{rank}(X) = r$, $r \leq n$.

This note is in five parts: the first part describes how the data vector y contributes to the inference about the vector of marker effects β in this particular setup where the likelihood depends on β only through $X\beta$. The basic idea is to express the vector of marker effects as the sum of two orthogonal linear combinations of the elements of β: one that is identifiable and the other that is unidentifiable by the data. The second part examines the process of Bayesian learning in the genomic linear model (1) under two scenarios: one where variance components are known and the other where they are jointly inferred with the genetic marker effects β. The third, fourth and fifth parts discuss how identifiability of the vector of marker effects plays out in the context of the classical ridge regression, the lasso and the spike and slab model, respectively.

Preliminaries

1. Use will be made of the Moore-Penrose inverse (Searle, 1982) although this is not mandatory. Given a matrix X, the Moore-Penrose inverse X^+ is a *unique* matrix that satisfies the following criteria:

   - $XX^+X = X$

- $X^+XX^+ = X^+$
- XX^+ is symmetric
- X^+X is symmetric

Given X, there are many matrices G that satisfy the first of these conditions. This is in contrast with the Moore-Penrose inverse that is unique.

2. Singular value decomposition. In general, if X is an $(n \times m)$ matrix of rank r, then it can be written as

$$X = UDV', \tag{2}$$

where

- U is an $n \times r$ matrix whose columns are orthonormal left singular vectors of X;
- $D = \text{diag}(d_1, \ldots, d_r)$ is an $r \times r$ diagonal matrix whose diagonal elements satisfy $d_1 \geq d_2 \geq \cdots \geq d_r > 0$ and are the positive singular values of X;
- V is an $m \times r$ matrix whose columns are orthonormal right singular vectors of X.

The columns of U and V are orthonormal, that is, $U'U = I_r$, $V'V = I_r$, and unless $r = n$, $UU' \neq I_n$; rather, UU' is the orthogonal projector onto the column space of X. Similarly, unless $r = m$, $VV' \neq I_m$; instead, VV' is the orthogonal projector onto the row space of X.

3. Define the $m \times m$ projection matrix P_R as

$$\begin{aligned} P_R &= X'(XX')^+X \\ &= X^+X \end{aligned} \tag{3}$$

where $(XX')^+$ is the Moore-Penrose inverse of XX' and X^+ is the Moore-Penrose inverse of X. Matrix P_R is the orthogonal projector of vectors onto the row space of matrix X. This means that $\beta_R = P_R\beta$ lies in the row space of X, whereas $\beta_N = (I - P_R)\beta$ lies in the null space of X. Then,

$$\begin{aligned} X\beta_N &= X(I - P_R)\beta \\ &= X\beta - XX^+X\beta \\ &= 0 \end{aligned} \tag{4}$$

because $XX^+X = X$, due to the first Penrose identity.

The proof of equality (3) is as follows. From the partitioning (2),

$$XX' = UDV'VDU' = UD^2U' \Rightarrow (XX')^+ = UD^{-2}U',$$

therefore

$$X'(XX')^{+}X = VDU'(UD^{-2}U')UDV' = VV'.$$

Inverting (2)

$$X^{+} = VD^{-1}U' \Rightarrow X^{+}X = VD^{-1}U'UDV' = VV',$$

establishing the equality. In this expression, $D^{-1} = \text{diag}\left(\frac{1}{d_1}, \ldots, \frac{1}{d_r}\right)$, where $r = \text{rank}(X)$.

When the projection matrix is computed with the Moore-Penrose inverse, $P_R = P_R^2$ and $P_R' = P_R$; that is, P_R is idempotent and symmetric. However, the Moore–Penrose inverse is not required in the construction of these orthogonal projectors. Any generalized inverse satisfying $AGA = A$, with $A = XX'$ for P_R or $A = X'X$ for P_C, produces the same unique orthogonal projector.

Partitioning the vector of genetic marker effects

The vector β can be expressed as the sum of two orthogonal components

$$\begin{aligned} \beta &= P_R\beta + (I - P_R)\beta \\ &= \beta_R + \beta_N, \end{aligned} \tag{5}$$

where $\beta_R'\beta_N = 0$ (NOTE: $\beta_R'\beta_N = \beta'P_R'(I - P_R)\beta = \beta'0\beta = 0$, using $P_R = P_R'$, $P_R^2 = P_R$). Then,

$$\begin{aligned} X\beta &= X\beta_R + X\beta_N \\ &= X\beta_R \end{aligned}$$

due to (4). Therefore the model $y = X\beta + e$ can be written as $y = X\beta_R + e$: data y contain information about the row space component, β_R, but not about the null space component, β_N. When the likelihood depends on β only through $X\beta$, the fraction of β that lies in the null space of X, $(I - P_R)\beta$, is not identifiable from y.

When the rank of X is $r = m$, $VV' = I_m$. In this case $P_R = I_m$, $(I_m - P_R) = 0$, and $\beta_N = 0$, indicating that the null space has disappeared; the entire vector β is identifiable from the data. Then

$$P_R\beta = \beta.$$

Parametrising the genomic model in terms of genomic values

Assume that instead of parametrising the genomic model in terms of marker effects β, the model takes the form:

$$y \mid g, \sigma^2 \sim N\left(g,\, I\sigma^2\right), \tag{6a}$$

$$g \mid \sigma_G^2 \sim N\left(0,\, G\sigma_G^2\right) \tag{6b}$$

where $G = \frac{XX'}{m}$ and $\sigma^2_G = m\sigma^2_\beta$. The covariance matrix $G\sigma^2_G$ is positive semidefinite (possibly singular), so g has a singular multivariate normal distribution whose support is the column space of X, whenever $\text{rank}(X) < n$, $m > n$. Despite the rank deficiency of X, under this parametrisation, every admissible value of the parameter g is identifiable, as shown below.

Define $P_C = XX^+$, the symmetric and idempotent $n \times n$ matrix that projects vectors onto the column space of X. Notice that this is in contrast with the $m \times m$ matrix P_R, a projector onto the row space of X. Then, with $g = X\beta$,

$$\begin{aligned} P_C\, g &= XX^+X\beta \\ &= X\beta = g \end{aligned}$$

because $XX^+X = X$, due to the first Penrose identity. Every admissible parameter g lies in the column space of X; thus the parameter space coincides with the identifiable parameter space and the projector P_C is the identity operator. There is no unidentifiable component of g, in contrast with the partitioning in (5): for β, the parameter space is $\mathbb{R}^m$ but only the projection onto the row space of X is identifiable.

The second part of this note examines how identifiability of β plays out when matrix X of the genomic model (1) is also rank deficient but variance components σ^2_β and σ^2 are unknown and updated by the data.

Bayesian learning: updating genetic marker effects

The question addressed is whether, given a hierarchical Bayesian model and a body of data, Bayesian inference leads to learning about all components of vector β. This is investigated by comparing the prior and posterior distributions of the marker effects. If the posterior distribution of a component is identical to its prior distribution, the data provide no information about that component beyond that already contained in the prior, and no Bayesian learning has occurred for that component.

Variance components known

Consider again the standard genomic model that is described by the following hierarchy:

$$y \mid \beta, \sigma^2 \sim N\big(X\beta,\, I\sigma^2\big)\,, \tag{7a}$$

$$\beta \mid \sigma^2_\beta \sim N\big(0,\, I\sigma^2_\beta\big)\,. \tag{7b}$$

The vector y contains the n phenotypic observations, X is the observed $(n \times m)$ marker matrix with $m > n$ and rank $r < \min(n, m)$, and β is the unobserved $(m \times 1)$ vector of marker effects. The variance components σ^2 and σ^2_β are assumed known.

As indicated in the first part of this note, the vector of genetic marker effects can be partitioned into a component that is identifiable from the data, β_R, and an orthogonal component that is not identifiable from the data, β_N. The likelihood for the genomic

model can be written as $p(y|\beta_R, \beta_N, \sigma^2) = p(y|\beta_R, \sigma^2)$ because $X\beta_N = 0$, and therefore the posterior distribution of β takes the following form:

$$p\left(\beta_R, \beta_N | y, \sigma^2, \sigma_\beta^2\right) \propto p(y|\beta_R, \sigma^2) p\left(\beta|\sigma_\beta^2\right).$$

As shown in the NOTE below, given the prior in (7b), the prior density of vector β factorises as $p\left(\beta|\sigma_\beta^2\right) = p\left(\beta_R|\sigma_\beta^2\right) p\left(\beta_N|\sigma_\beta^2\right)$; therefore, since the likelihood does not include β_N, the posterior distribution of β_N, after integrating over β_R, is given by

$$p\left(\beta_N | y, \sigma^2, \sigma_\beta^2\right) \propto p\left(\beta_N|\sigma_\beta^2\right),$$

indicating absence of Bayesian learning for the non-identifiable component β_N.

NOTE

Under normality, with $\beta|\sigma_\beta^2 \sim N(0, I_m\sigma_\beta^2)$ and $\beta_R = P_R\beta$, $\beta_N = (I_m - P_R)\beta$,

$$\begin{aligned} Cov(\beta_R, \beta'_N|\sigma_\beta^2) &= Cov(P_R\beta, \beta'(I_m - P_R)|\sigma_\beta^2) \\ &= P_R(I_m - P_R)\sigma_\beta^2 = 0 \end{aligned}$$

since P_R is symmetric and idempotent. Therefore $p\left(\beta|\sigma_\beta^2\right) = p\left(\beta_R|\sigma_\beta^2\right) p\left(\beta_N|\sigma_\beta^2\right)$.

Variance components unknown

Assume that σ_β^2 is unknown and updated from the data, and for simplicity, assume σ^2 is known. The posterior distribution of β_N takes the following form:

$$p\left(\beta_N|y, \sigma^2\right) = \int p\left(\beta_N|y, \sigma_\beta^2, \sigma^2\right) p\left(\sigma_\beta^2|y\right) d\sigma_\beta^2.$$

Since $p\left(\beta_N|y, \sigma_\beta^2, \sigma^2\right) = p\left(\beta_N|\sigma_\beta^2\right)$, the posterior distribution becomes

$$p\left(\beta_N|y, \sigma^2\right) = \int p\left(\beta_N|\sigma_\beta^2\right) p\left(\sigma_\beta^2|y, \sigma^2\right) d\sigma_\beta^2.$$

This expression is different from

$$\int p\left(\beta_N|\sigma_\beta^2\right) p\left(\sigma_\beta^2\right) d\sigma_\beta^2$$

unless $p\left(\sigma_\beta^2|y, \sigma^2\right) = p\left(\sigma_\beta^2\right)$ which is usually not the case since the data updates σ_β^2. However, the mean of the posterior distribution of β_N remains unchanged relative to the prior mean. Indeed,

$$\int\int \beta_N\, p\left(\beta_N|\sigma_\beta^2\right) p\left(\sigma_\beta^2|y, \sigma^2\right) d\beta_N d\sigma_\beta^2 =$$
$$\int \left[\int \beta_N\, p\left(\beta_N|\sigma_\beta^2\right) d\beta_N\right] p\left(\sigma_\beta^2|y, \sigma^2\right) d\sigma_\beta^2 = 0$$

since the integral in square brackets corresponds to the prior mean.

Conclusion: when β and σ^2_β are inferred from a joint Bayesian analysis, the nonidentifiable component β_N of β undergoes indirect Bayesian learning through the posterior updating of the variance component σ^2_β. The likelihood remains completely flat in the β_N direction, so the identifiability structure of the model is unchanged: β_N is still not directly identifiable from the data. These results and those presented in the first part of this note, highlight the distinction between inference of genomic values $g = X\beta$ and inference of marker effects β. Genomic values are identifiable whereas only the β_R component of β is learned directly from the data. Consequently in genomic models where $m \gg n$, inference of individual marker effects is inherently more challenging than inference of genomic values.

How Does the Penalty Term in Ridge Regression Alter the Identifiability of β?

The first part of this note discussed identifiability of the parameter β of the linear regression model $y = X\beta + e$ when matrix X is rank deficient. It was indicated that the vector of genetic marker effects β can be partitioned into two orthogonal linear combinations: β_R that is identifiable by the data y, and β_N that is not (see page 3). The second part of this note shows how a Bayesian implementation deals with the rank deficiency of matrix X in order to infer β. When variance components are known and the only unknown parameter is β, given a Gaussian prior distribution for β of the form $N(0, I\sigma^2_\beta)$, the Bayesian analysis updates β_R via the data, but the information to infer the unidentifiable component β_N originates solely from the prior distribution. The posterior mode of β_N (and given the Gaussian assumption, the posterior mean) is equal to zero. This third part illustrates how this process unfolds in the context of ridge regression.

Referring to page 3, partition β into its two orthogonal components:

$$\beta = \beta_R + \beta_N. \tag{8}$$

The ridge coefficients are obtained as

$$\hat{\beta} \underset{\arg\min\ \beta}{=} (y - X\beta)'(y - X\beta) + \lambda\beta'\beta,$$

where here, the shrinkage parameter $\lambda > 0$. Using (8), the orthogonality of β_R and β_N, and the equality $X\beta = X\beta_R$, the cost function of the ridge regression is

$$J(\beta_R, \beta_N|\lambda) = (y - X\beta_R)'(y - X\beta_R) + \lambda\beta'_R\beta_R + \lambda\beta'_N\beta_N.$$

For any fixed value of β_R, the first two terms are constant with respect to β_N. Therefore

$$\hat{\beta}_N \underset{\arg\min\ \beta_N}{=} \lambda\,\beta'_N\beta_N.$$

Since $\lambda > 0$, $\lambda\,\beta'_N\beta_N \geq 0$, with equality only if $\beta_N = 0$. Thus,

$$\hat{\beta}_N = 0. \tag{9}$$

The remaining component $\hat{b}_R$ is obtained by minimising

$$(y - X\beta_R)'(y - X\beta_R) + \lambda\,\beta_R'\beta_R.$$

Conclusion: Ridge regression only estimates the component of β that is identifiable by the data: the data determine only β_R. The unidentifiable component β_N is set to zero because this minimises the cost function: the penalty determines β_N. The solution for β_N agrees with the Bayesian approach under the Gaussian prior for β: the posterior distribution of β_N is equal to its prior distribution, whose mean (and mode) is zero.

How Does the Penalty Term in Lasso Alter the Identifiability of β?

This section summarises the concept of identifiability of genetic marker effects β, when the matrix of observed marker effects X is rank deficient and lasso is used as the operational model. As before, partition vector β into its orthogonal components

$$\beta = \beta_R + \beta_N. \tag{10}$$

Then, since b_N belongs to the null space of X, $X\beta = X\beta_R$, and the loss function associated with lasso can be written as

$$J(\beta_R, \beta_N|\lambda) = (y - \beta_R)'(y - X\beta_R) + \lambda\ \|\beta_R + \beta_N\|_1 \tag{11}$$

where the last term in the right hand side is the lasso's ℓ_1-norm penalty. In the case of ridge regression, $\|\beta_R + \beta_N\|_2^2 = \|\beta_R\|_2^2 + \|\beta_N\|_2^2$, which upon minimisation leads immediately to $\hat{\beta}_N = 0$, regardless of the value of β_R. However, this factorisation does not hold for the last term in (11) despite the orthogonality of β_R and β_N. Therefore, for a fixed value of β_R, minimisation of the loss function (11) with respect to β_N, where β_N belongs to the null space of X, does not necessarily imply $\hat{\beta}_N = 0$.

Conclusion: The data contain no direct information about the null-space component β_N. Conditional on β_R, the value of β_N is therefore entirely determined by the ℓ_1 penalty. Nevertheless, because the estimated β_R depends on the data, the lasso selected value of β_N can depend indirectly on the data through β_R.

Inference of marker effects using the spike and slab model when the matrix of marker genotypes is rank deficient

This subsection elaborates on the subject of Bayesian learning in the case of the spike and slab model.

The first point to be made is that the lack of direct identifiability of β via the data also holds for the Bayesian spike and slab model. However, in common with the Bayesian linear genomic model where β, a priori, is assigned a $N(0, I\sigma_\beta^2)$, the spike and slab model provides an avenue of Bayesian learning through the posterior association between β and

other parameters of the hierarchical model, about which the data may provide information. This source of indirect identifiability is arguably richer in the case of the spike and slab model because the number of contributing parameters is larger and this creates more routes for indirect learning. For example, consider the unknown parameter π, the proportion of markers included. Although the likelihood does not identify the m-dimensional parameter β, it may provide information about π. Updating π updates $\Pr(\delta_j = 1|y)$ which changes the posterior distribution of β_j. Thus, there is the path

$$y \to \pi \to \delta_j \to \beta_j.$$

Likewise, learning about σ^2_β changes the distribution of the non-zero effects:

$$y \to \sigma^2_\beta \to \beta_j \mid \delta_j = 1.$$

So the relatively rich parameterisation is not merely an additional identifiability burden. It also supplies additional hierarchical pathways through which information can reach β.

An important mechanism peculiar to the spike and slab model

Beyond these hierarchical pathways of Bayesian learning, the spike-and-slab model offers another mechanism for addressing rank deficiency: it assigns positive prior probability to submodels containing only a subset of the available markers.

Under the usual setup where X is rank deficient and only a very small fraction of the m marker genotypes have an effect on the phenotype, the spike-and-slab prior can favour sparse solutions among the vectors β producing the same $X\beta$. Consequently the posterior can become concentrated on one or several sparse representations of $X\beta$. In an McMC environment, each round of the Markov chain visits a particular reduced model/configuration, defined by the realised values of the vector containing the m binary indicators δ, of which a small fraction take the value 1. Within each configuration, if the resulting matrix X_M is of full column rank, the associated β_M becomes identifiable, conditional on the configuration. The McMC sampler averages parameters over the posterior uncertainty of the different configurations without explicitly enumerating the configurations or calculating their individual posterior probabilities. Indeed, for marker effect β_i, say, the model-averaged posterior mean is

$$\mathrm{E}(\beta_i \mid y) = \sum_{k:i\in M_k} \mathrm{E}(\beta_i \mid y, M_k) \Pr(M_k \mid y).$$

The posterior expectation of the effect of marker i is obtained by averaging its conditional posterior expectations over all configurations that include the marker, weighting each expectation by the posterior probability of the corresponding configuration. Configurations excluding marker i make no contribution because its effect is then zero.

After convergence, assuming adequate mixing of the Markov chain, the sampler visits its model configurations according to their posterior probabilities and the Monte Carlo

averages implicitly perform the required weighting. For example,

$$\hat{\mathrm{E}}(\beta_i \mid y) = \frac{1}{T}\sum_{j=1}^{T} \beta_i^{[j]}$$

that is the Monte Carlo approximation to the Bayesian model-averaged posterior mean.

Marker inclusion probabilities in the spike and slab model

The same principle of Bayesian model averaging applies to the inclusion indicators, whose posterior expectations have a particularly simple interpretation.

In the spike and slab model, the posterior inclusion probability $\Pr(\delta = 1 \mid y)$ of a marker is obtained by summing over all the configurations that contain the marker. To see this, consider the following schematic representation of the spike and slab model,

$$\beta_i = \alpha_i \delta_i; \quad \alpha_i | \sigma_\beta^2 \sim N(0, \sigma_\beta^2);\ \delta_i \sim \text{Bernoulli}(\pi),$$

where α_i and δ_i are independent a priori conditional on hyperparameters. A configuration M defines a particular submodel,

$$y = X_M \beta_M + e.$$

Within this submodel, the value of δ_j is known:

$$\delta_j\,(M) = \begin{cases} 1,\, j \in M, \\ 0,\, j \notin M. \end{cases}$$

Consequently,

$$\begin{aligned} \mathrm{E}(\delta_j \mid y) &= \sum_M \delta_j(M) \Pr(M \mid y) \\ &= \sum_{M: j \in M} \Pr(M \mid y), \end{aligned}$$

and since δ_j is binary,

$$\mathrm{E}(\delta_j \mid y) = \Pr(\delta_j = 1 \mid y).$$

Therefore the posterior inclusion probability of marker j is obtained by summing over all the posterior probabilities of the configurations that include marker j.

Connection with McMC

In practice, when the number of markers m is large the 2^m configurations cannot be enumerated. Instead, the sampler visits configurations $M_1, M_2, \ldots, M_T$. At iteration t,

$$\delta_j^{[t]} = \begin{cases} 1,\, j \in M^{[t]}, \\ 0,\, j \notin M^{[t]}. \end{cases}$$

Hence,

$$\hat{\Pr}(\delta_j = 1 \mid y) = \frac{1}{T}\sum_{t=1}^{T} \delta_j^{[t]}.$$

This estimates the sum of posterior probabilities over all configurations containing marker j, without explicitly computing the posterior probability of each configuration.

Conclusion: In a Bayesian spike and slab analysis, McMC sampling implicitly performs Bayesian model averaging over the possible configurations of included markers. Although the number of configurations may be prohibitively large for explicit enumeration, posterior expectations and predictions can be approximated by averaging over the configurations visited by the sampler. Many of these configurations may correspond to marker matrices of full column rank, so that the associated marker effects are identifiable conditional on the selected submodel. Thus, by favouring sparse configurations, the spike and slab model can mitigate the identifiability problem associated with rank deficiency. However, different configurations may provide equally plausible explanations of the data, and identifiability within individual configurations does not imply that the configurations themselves can be distinguished. Bayesian model averaging accounts for this uncertainty without necessarily resolving the lack of identifiability of the complete vector of marker effects through the likelihood. Nevertheless, the hierarchical structure of the spike and slab model provides additional pathways for Bayesian learning about marker effects through their posterior association with other model parameters.

An additional consideration arises when genotypes are correlated, particularly in the context of gene discovery. By fitting marker effects jointly, the spike and slab model can distinguish nonzero conditional effects from marginal associations induced by linkage disequilibrium. However, when several sparse configurations provide similar explanations of the data, uncertainty remains about which particular markers have nonzero effects. This uncertainty is reflected in the marginal posterior inclusion probabilities, which may be relatively small for individual markers even when the posterior probability that at least one marker in a group has an effect is close to one. Thus, while Bayesian model averaging can be advantageous for prediction, where different configurations may yield similar predicted values, identifying individual markers with nonzero effects may remain difficult, especially in regions of strong linkage disequilibrium.

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